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Hardware

GIGA Display Shield



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GIGA Display Shield Camera Connector Guide

Learn how to use the camera connector with the GIGA Display Shield.

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Introduction

The GIGA Display Shield comes with an Arducam® camera connector. In this tutorial, we will go through what cameras are compatible with the display shield, how to connect the camera, and how to run a sketch to stream the camera feed to the display.

Hardware & Software Needed

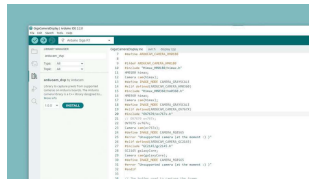
[Arduino GIGA R1 WiFi](#)[Arduino GIGA Display Shield](#)[Arduino IDE](#)

HM01B0, HM0360, GC2145 or OV7675 camera

Downloading the Library and Board Package

[Help](#)

The Arduino Mbed OS Giga Boards Board Package contains most of the libraries you need to work with the shield's camera connector. To install the Board Package for GIGA boards, navigate to **Tools > Board > Boards Manager** or click the Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for giga and install the latest Arduino Mbed OS Giga Boards version, the [Arduino_H7_Video library](#) library is included in the Board Package. Now open the library tab on the left, search for [Arducam_dvp](#), and install this library.



Library tab in the Arduino IDE

Compatible Cameras

The GIGA Display Shield is compatible with the following cameras:

[HM01B0](#)

[HM0360](#)

[GC2145](#)

[OV7675](#)

Connect the camera to the connector on the front of the display shield as shown in the image below.

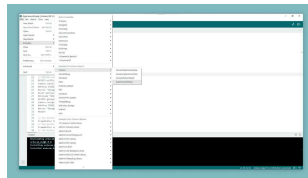


Camera connected to the GIGA Display Shield

Note the direction of the shield, physically the connector can be plugged in from both directions, but it will only work when facing the same direction as the display.

Full Sketch

Open the example sketch by going to **File > Examples > Camera > GigaCameraDisplay** in the Arduino IDE.



Opening the example sketch in the Arduino IDE

Whichever of the compatible cameras you are using the sketch will include libraries and definitions for them all, meaning only one line needs to be changed depending on what camera you will be using. The line that needs to be changed is this one:

```
1 #define ARDUCAM_CAMERA_HM
```

Depending on what camera you are using it should be changed accordingly:

LM01DP

HM0360:

```
#define
ARDUCAM_CAMERA_HM0360
```

GC2145:

```
#define
ARDUCAM_CAMERA_GC2145
```

OV7675:

```
#define
ARDUCAM_CAMERA_OV767x
```

The sketch will then capture frames into the framebuffer and print a live camera feed to the display.

 Note: make sure to connect/disconnect the camera when the board is powered off.

```
1 #include "arducam_dvp.
2 #include "Arduino_H7_v
3 #include "dsi.h"
4 #include "SDRAM.h"
5
6 // This example only w
7 #define ARDUCAM_CAMERA
8
9 #ifdef ARDUCAM_CAMERA_
10 #include "Himax_HM01B0
11 HM01B0 himax;
12 Camera cam(himax);
13 #define IMAGE_MODE CAM
14 #elif defined(ARDUCAM_
15 #include "Himax_HM0360
16 HM0360 himax;
17 Camera cam(himax);
18 #define IMAGE_MODE CAM
19 #elif defined(ARDUCAM_
20 #include "OV7670/ov767
21 // OV7670 ov767x;
22 OV7675 ov767x;
23 Camera cam(ov767x);
24 #define IMAGE_MODE CAM
25 #elif defined(ARDUCAM_
26 #include "GC2145/gc214
27 GC2145 galaxyCore;
28 Camera cam(galaxyCore)
29 #endif
30 #define IMAGE_MODE CAM
```



Note: Depending on the camera being used the captured resolution might be smaller than the screen. If this is the case, there will be a black bar displayed next to the captured feed, it will not be automatically centered.

Conclusion

This tutorial went through how to connect a compatible camera to the shield and also how to test it out quickly with the example sketch included in the Board Package. Now you should see a live feed from the camera on your GIGA Display Shield!

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